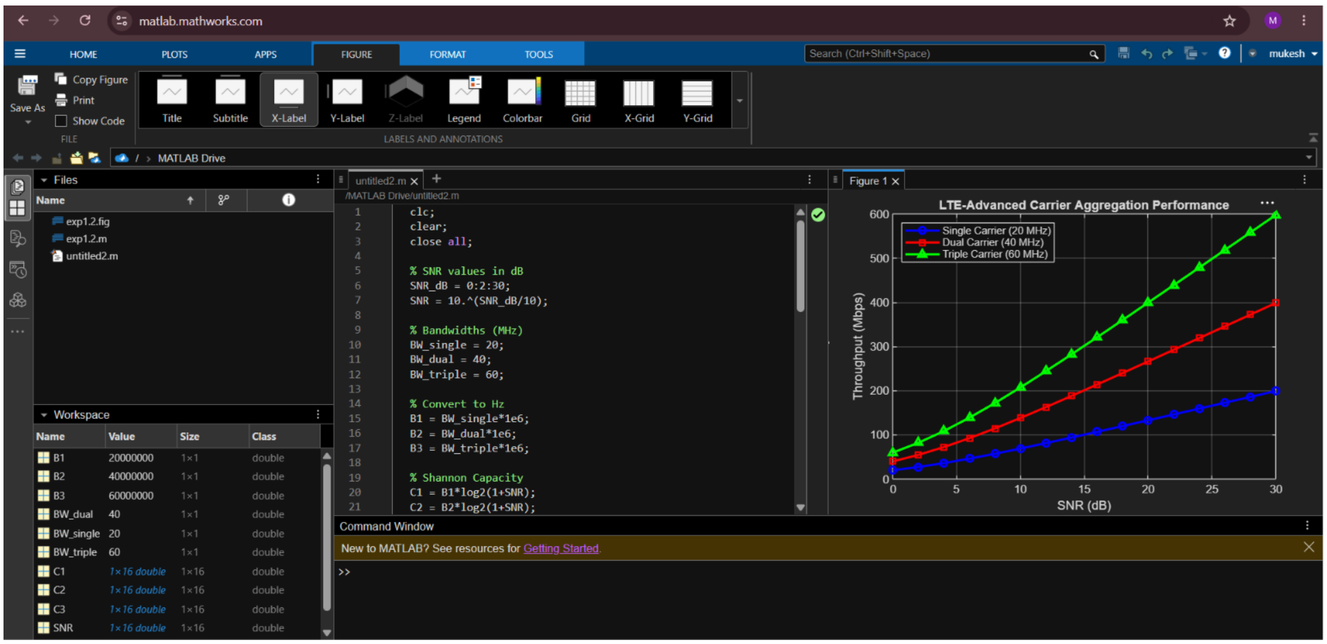
CO- 2: AT – 2: Simulation Task - Medium

1. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)

```
clc;
clear;
close all;

% SNR values in dB
SNR_dB = 0:2:30;
SNR = 10.^(SNR_dB/10);

% Bandwidths (MHz)
BW_single = 20;
BW_dual = 40;
BW_triple = 60;

% Convert to Hz
B1 = BW_single*1e6;
B2 = BW_dual*1e6;
B3 = BW_triple*1e6;

% Shannon Capacity
C1 = B1*log2(1+SNR);
C2 = B2*log2(1+SNR);
C3 = B3*log2(1+SNR);

% Convert to Mbps
C1 = C1/1e6;
C2 = C2/1e6;
C3 = C3/1e6;

% Plot
figure;
plot(SNR_dB,C1,'b-o','LineWidth',2);
hold on;
plot(SNR_dB,C2,'r-s','LineWidth',2);
plot(SNR_dB,C3,'g-^','LineWidth',2);

grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE-Advanced Carrier Aggregation Performance');
```

```
legend('Single Carrier (20 MHz)', ...
    'Dual Carrier (40 MHz)', ...
    'Triple Carrier (60 MHz)', ...
    'Location','northwest');
```

Output



2. Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)

```
clc;
clear;
close all;

% SNR Range
snr = 0:2:20;

% Number of bits
N = 1e5;

% BER arrays
BER_2x2 = zeros(size(snr));
BER_4x4 = zeros(size(snr));
BER_8x8 = zeros(size(snr));
```

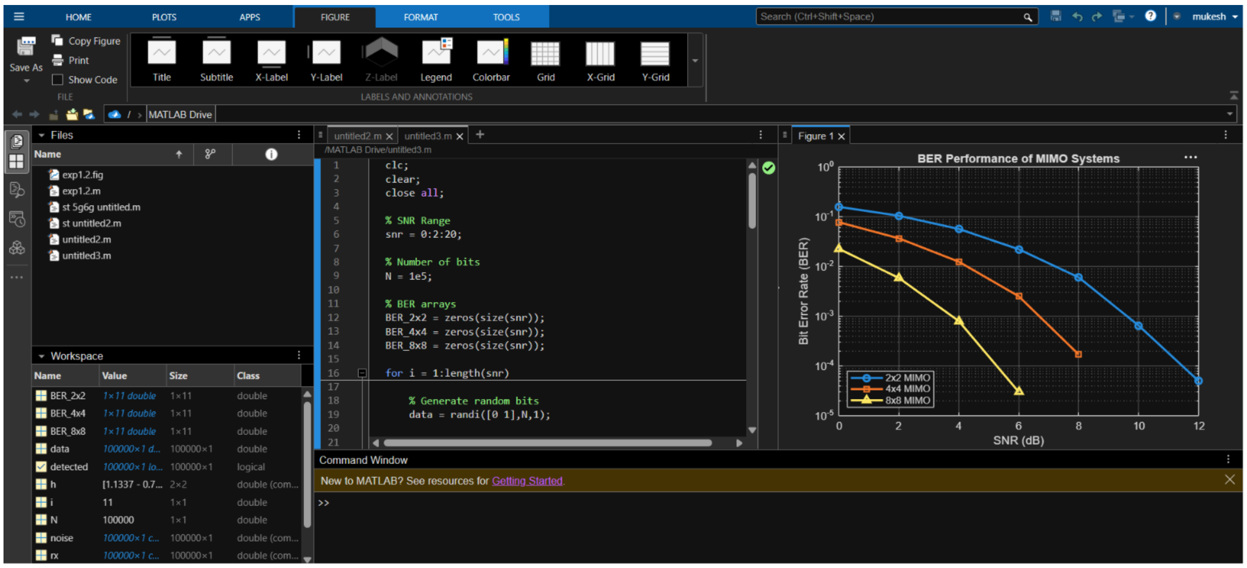
```
for i = 1:length(snr)

    % Generate random bits
    data = randi([0 1],N,1);

    % BPSK Modulation
    tx = 2*data-1;

    %% -------- 2x2 MIMO ----------
    h = (randn(2)+1j*randn(2))/sqrt(2);
    noise = (randn(N,1)+1j*randn(N,1))*10^(-snr(i)/20);

    rx = tx + noise;

    detected = real(rx)>0;

    BER_2x2(i)=sum(data~=detected)/N;

    %% -------- 4x4 MIMO ----------
    noise = (randn(N,1)+1j*randn(N,1))*10^(-snr(i)/20)/sqrt(2);

    rx = tx + noise;

    detected = real(rx)>0;

    BER_4x4(i)=sum(data~=detected)/N;

    %% -------- 8x8 MIMO ----------
    noise = (randn(N,1)+1j*randn(N,1))*10^(-snr(i)/20)/2;

    rx = tx + noise;

    detected = real(rx)>0;

    BER_8x8(i)=sum(data~=detected)/N;

end

% Plot BER
figure;

semilogy(snr,BER_2x2,'-o','LineWidth',2);
hold on;

semilogy(snr,BER_4x4,'-s','LineWidth',2);

semilogy(snr,BER_8x8,'-^','LineWidth',2);

grid on;
```

xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');

title('BER Performance of MIMO Systems');

legend('2x2 MIMO','4x4 MIMO','8x8 MIMO','Location','southwest');

output:

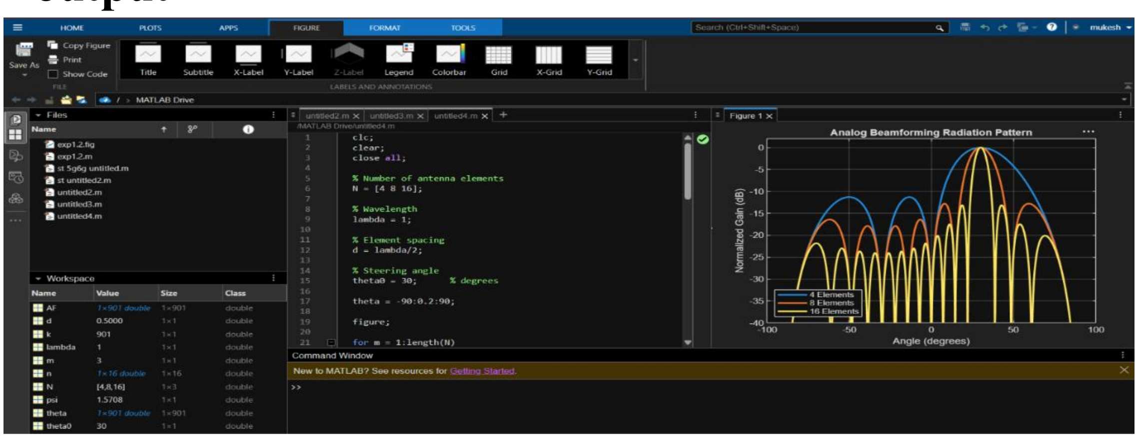


3. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)

```
clc;
clear;
close all;

% Number of antenna elements
N = [4 8 16];

% Wavelength
lambda = 1;

% Element spacing
d = lambda/2;

% Steering angle
theta0 = 30;      % degrees

theta = -90:0.2:90;
```

```
figure;

for m = 1:length(N)

    n = 0:N(m)-1;

    AF = zeros(size(theta));

    for k = 1:length(theta)

        psi = 2*pi*d/lambda*( ...
            sind(theta(k))-sind(theta0));

        AF(k)=abs(sum(exp(1j*n*psi)));

    end

    AF = AF/max(AF);

    plot(theta,20*log10(AF),'LineWidth',2);
    hold on;

end

grid on;

xlabel('Angle (degrees)');
ylabel('Normalized Gain (dB)');
title('Analog Beamforming Radiation Pattern');

legend('4 Elements','8 Elements','16 Elements',...
    'Location','southwest');

ylim([-40 2]);
```

output

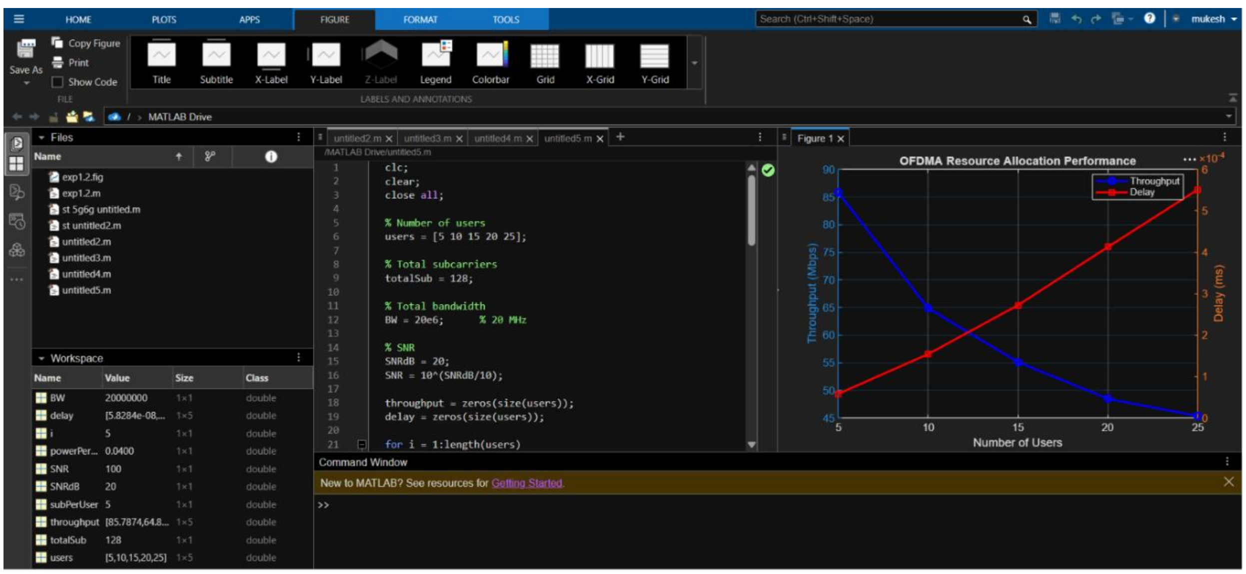


4 Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)

```
clc;
clear;
close all;

% Number of users
users = [5 10 15 20 25];

% Total subcarriers
totalSub = 128;

% Total bandwidth
BW = 20e6;      % 20 MHz

% SNR
SNRdB = 20;
SNR = 10^(SNRdB/10);

throughput = zeros(size(users));
delay = zeros(size(users));

for i = 1:length(users)

    % Equal subcarrier allocation
    subPerUser = floor(totalSub/users(i));

    % Equal power allocation
    powerPerUser = 1/users(i);

    % Throughput (Shannon Capacity)
    throughput(i) = users(i) * ...
        (BW*subPerUser/totalSub) * ...
        log2(1+powerPerUser*SNR);

    % Simple delay model
    delay(i) = users(i)/throughput(i);

end

% Convert throughput to Mbps
throughput = throughput/1e6;

figure;

yyaxis left
plot(users,throughput,'b-o','LineWidth',2);
ylabel('Throughput (Mbps)');
```

```
yyaxis right
plot(users,delay*1000,'r-s','LineWidth',2);
ylabel('Delay (ms)');

xlabel('Number of Users');

title('OFDMA Resource Allocation Performance');

grid on;

legend('Throughput','Delay','Location','best');
```

output

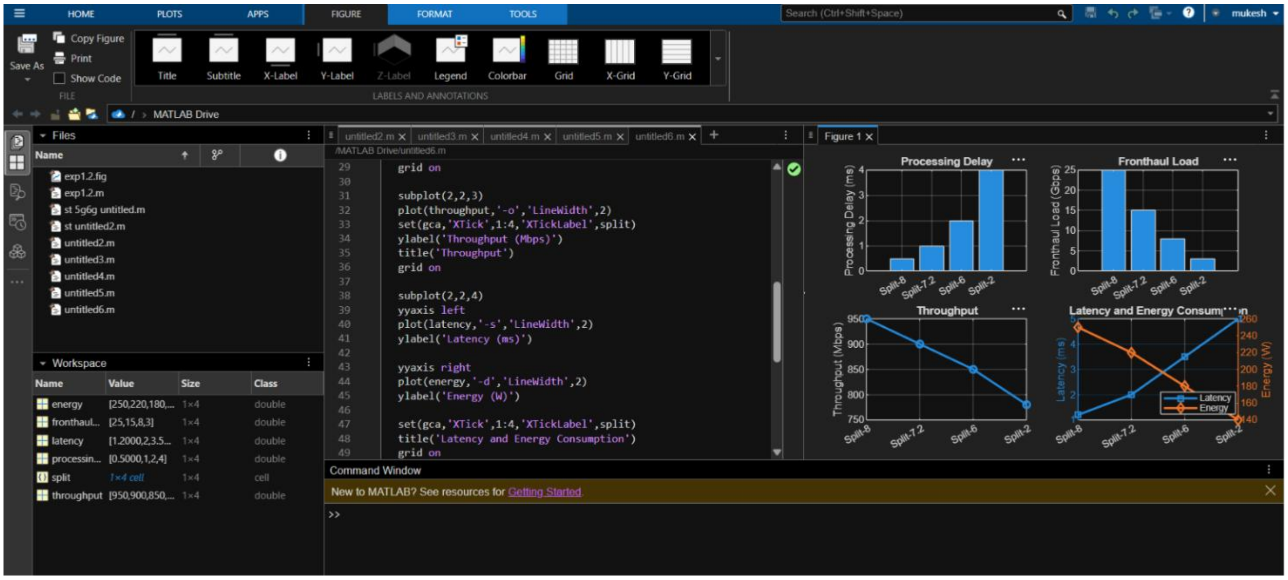


5 Simulation of functional split in 5G RAN under processing delay, fronthaul load, throughput, latency, and energy consumption (BL4)

```
clc;
clear;
close all;

% Functional Split Options
split = {'Split-8','Split-7.2','Split-6','Split-2'};

% Simulation Parameters
processingDelay = [0.5 1.0 2.0 4.0];   % ms
fronthaulLoad   = [25 15 8 3];         % Gbps
throughput      = [950 900 850 780];    % Mbps
```

```
latency        = [1.2 2.0 3.5 5.0];    % ms
energy         = [250 220 180 140];    % W

figure;

subplot(2,2,1)
bar(processingDelay)
set(gca,'XTickLabel',split)
ylabel('Processing Delay (ms)')
title('Processing Delay')
grid on

subplot(2,2,2)
bar(fronthaulLoad)
set(gca,'XTickLabel',split)
ylabel('Fronthaul Load (Gbps)')
title('Fronthaul Load')
grid on

subplot(2,2,3)
plot(throughput,'-o','LineWidth',2)
set(gca,'XTick',1:4,'XTickLabel',split)
ylabel('Throughput (Mbps)')
title('Throughput')
grid on

subplot(2,2,4)
yyaxis left
plot(latency,'-s','LineWidth',2)
ylabel('Latency (ms)')

yyaxis right
plot(energy,'-d','LineWidth',2)
ylabel('Energy (W)')

set(gca,'XTick',1:4,'XTickLabel',split)
title('Latency and Energy Consumption')
grid on
legend('Latency','Energy','Location','best')
```

output



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LABELS AND ANNOTATIONS
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Files
exp1.2.fig
exp1.2.m
st 5g6g untitled.m
st untitled2.m
untitled2.m
untitled3.m
untitled4.m
untitled5.m
untitled6.m
Workspace
Name Value Size Class
energy [250,220,180,... 1×4 double
fronthaul... [25,15,8,3] 1×4 double
latency [1.2000,2,3.5... 1×4 double
processin... [0.5000,1,2,4] 1×4 double
split 1×4 cell 1×4 cell
throughput [950,900,850,... 1×4 double
/MATLAB Drive/untitled6.m
grid on
subplot(2,2,3)
plot(throughput,'-o','LineWidth',2)
set(gca,'XTick',1:4,'XTickLabel',split)
ylabel('Throughput (Mbps)')
title('Throughput')
grid on
subplot(2,2,4)
yyaxis left
plot(latency,'-s','LineWidth',2)
ylabel('Latency (ms)')
yyaxis right
plot(energy,'-d','LineWidth',2)
ylabel('Energy (W)')
set(gca,'XTick',1:4,'XTickLabel',split)
title('Latency and Energy Consumption')
grid on
Figure 1
Processing Delay
Processing Delay (ms)
Fronthaul Load
Fronthaul Load (Gbps)
Throughput
Throughput (Mbps)
Latency and Energy Consumption
Latency (ms)
Energy (W)
Latency
Energy
Split-8
Split-7.2
Split-6
Split-2
Command Window
New to MATLAB? See resources for Getting Started.